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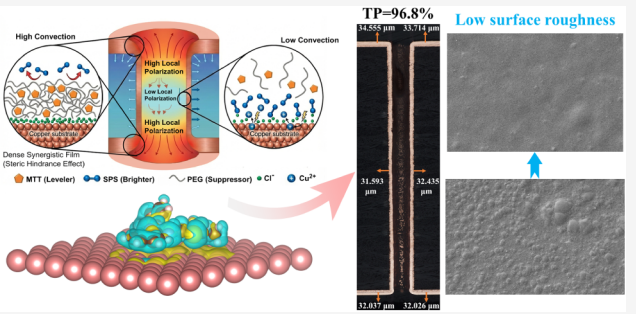
Thiazolyl Blue as a High-Efficiency Leveler for Conformal Copper Electrodeposition in High-Aspect-Ratio PCB Through-Holes

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Cite This: *Langmuir* 2026, 42, 15236−15247 Read Online

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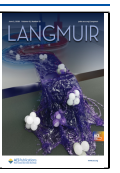
ABSTRACT: Driven by the rapid development of 5G/6G communications, high-performance computing, and artificial intelligence, the metallization of high-aspect-ratio through-holes (THs) in printed circuit boards (PCBs) imposes stringent requirements on the thickness uniformity and interconnect reliability of electroplated copper layers. This study systematically investigates the application of a novel leveler, thiazolyl blue (MTT), for PCB THs' electroplating, successfully realizing high-quality conformal deposition in high-aspect-ratio (10:1) THs. Electrochemical analyses demonstrate that MTT significantly enhances the cathodic polarization and effectively inhibits the reduction of Cu^{2+}. Meanwhile, microscopic morphological characterizations confirm that MTT exhibits excellent leveling capability, reducing the surface roughness of the copper deposit to as low as 0.147 μm. Furthermore, density functional theory (DFT) calculations reveal that the high inhibitory efficiency of MTT originates from the dominant interfacial adsorption of MTT at the cathode, with the bulk cation−π interaction with Cu^{2+} contributing as a minor secondary effect at the microscopic level. Crystallographic and interfacial wettability analyses further indicate that MTT effectively refines the grain size and promotes the preferential growth of the Cu(200) plane while improving the wettability of the plating bath. In practical high-aspect-ratio through-hole plating tests, the introduction of MTT substantially increases the throwing power (TP) from 25.2% to 96.8%. Based on these results, a competitive interfacial polarization remodeling mechanism governed by multiphysics coupling is proposed to elucidate the intrinsic role of MTT in promoting conformal through-hole deposition. This work provides a solid theoretical foundation and practical engineering guidance for the fabrication of highly reliable, high-frequency 3D interconnected PCB structures.



1. INTRODUCTION

Driven by the rapid advancement of 5G/6G communications, high-performance computing, and artificial intelligence (AI), printed circuit boards (PCBs)—the mother of electronic products—are undergoing profound technological revolutions.[1−3] As the fundamental carrier connecting electronic components, the PCB not only provides mechanical support for chips and passive devices but also functions as the neural network sustaining signal transmission and power delivery throughout the electronic system.[4] With the accelerated evolution of PCBs toward high integration, high interconnect density, and high-frequency/high-speed transmission, this trajectory poses two intertwined materials challenges. On the device side, the suppression of electromagnetic interference, interlayer crosstalk, and thermal accumulation in densely packed high-frequency boards has stimulated intensive research into advanced electromagnetic functional materials, with MXene-based two-dimensional absorbers and heterostructured composites emerging as a particularly promising family that delivers efficient GHz-to-THz electromagnetic-wave absorption and integrated thermal management.[5−8] On the interconnect side, the multilayer copper network that mechanically and electrically links the chips must, in parallel, be reformulated to simultaneously sustain signal integrity, ensure robust interconnect reliability, and guarantee long-term service lifespan under increasingly aggressive high-frequency operation—demands that ultimately translate into unprecedentedly stringent requirements on the copper metallization of microvias with high aspect ratios. Copper electrodeposition in high-aspect-ratio through-holes (THs) is the core technology for achieving interlayer electrical interconnection in multilayer PCBs. The quality of this metallization directly dictates the

Received: April 1, 2026
Revised: May 9, 2026
Accepted: May 11, 2026
Published: May 15, 2026



ACS Publications © 2026 American Chemical Society

https://doi.org/10.1021/acs.langmuir.6c01870
Langmuir 2026, 42, 15236−15247

signal integrity, current-carrying capacity, and long-term service lifespan of the electronic devices.[9,10]

However, during the practical electroplating of high-aspect-ratio THs, the initial current distribution is highly nonuniform due to geometric constraints. The local current density at the via opening is significantly higher than that at the center, accompanied by severe mass transfer limitations of Cu^{2+} within the hole. This multiphysics imbalance frequently triggers the "dog-bone" defect, characterized by excessive copper accumulation at the via opening and excessively thin deposits on the inner walls.[11] Such nonuniform deposition severely compromises the electrical and mechanical performance of the interconnected structures, constituting a universal bottleneck in the high-density interconnect (HDI) industry.[12] To overcome this challenge, it is imperative to introduce suppressors, accelerators, and levelers into the acidic copper sulfate bath.[13] The precise synergistic and competitive adsorption among these three additives dynamically modulates the deposition kinetics at different spatial locations within the TH, ultimately achieving high-quality conformal interconnection.

Typically, polyether macromolecules (e.g., polyethylene glycol, PEG) serve as suppressors; bridged by chloride ions, they profoundly enhance cathodic polarization and retard copper growth.[14] Sulfonate compounds bearing thiol or disulfide groups (e.g., bis(3-sulfopropyl) disulfide, SPS) act as accelerators: they capture Cu^{2+} at the cathode interface and lower the activation barrier for its reduction.[15] Among the plating formulations, the leveler plays the most critical interfacial regulatory role. By preferentially adsorbing at strongly convected regions (e.g., the opening), the leveler establishes a localized high-impedance barrier that inhibits overplating,[16−18] thereby compelling the spatial electric field lines to redistribute deeper into TH.[19−21] To date, various commercial dyes have been reported as levelers (e.g., Janus Green B, tetrazolium blue chloride, PAN).[22−24] Meanwhile, some researchers have devoted efforts to the synthesis of novel polymeric levelers (e.g., VIBDGE, PBDGE, Sudan-QNL).[25−27] Nevertheless, each of the aforementioned levelers exhibits specific limitations. Polymeric levelers such as VIBDGE and PBDGE, despite their promising performance, typically require multistep synthesis and tedious purification procedures, which considerably increase the production cost and hinder their practical scalability. Commercial dye-based levelers (e.g., Janus Green B) are readily accessible and have been widely adopted in industry with proven effectiveness; however, as the aspect ratio of through-holes continues to increase, further enhancement of throwing power remains highly desirable to ensure the thickness uniformity and long-term reliability of the copper interconnects. In this regard, MTT offers a compelling combination of advantages: it is a commercially available, off-the-shelf reagent that can be directly procured at low cost, entirely eliminating the need for custom synthesis or purification, and as will be demonstrated in Section 3.5, it delivers a throwing power (TP = 96.8%) that markedly outperforms the conventional leveler JGB (TP = 84.5%) under identical bath composition and process conditions. Therefore, a systematic investigation of MTT as a tailored leveler for high-aspect-ratio PCB through-hole electroplating is of urgent practical significance.

It is worth noting that MTT has previously been explored in other copper electrodeposition contexts. Ogutu et al. first introduced MTT as a single additive for superconformal filling of through-glass vias (TGVs) with aspect ratios up to 10:1 in methanesulfonate-based electrolytes.[28] More recently, Yang et al. compared MTT with iodonitrotetrazolium chloride (INT) in a sulfate-based acidic electroplating system for blind via copper filling and reported that MTT exhibits a conformal-type filling tendency rather than the bottom-up superfilling behavior required for microvia applications.[29] However, the performance of MTT in high-aspect-ratio PCB through-hole electroplating—an application scenario in which conformal, uniform thickness distribution rather than bottom-up superfilling is the primary objective—has not yet been systematically investigated. The present work therefore complements these prior studies by examining MTT as a leveler within a sulfate-based acidic electroplating system specifically tailored for through-hole plating, thereby establishing the suitability of MTT for this distinct application scenario. It should also be noted that our group recently reported tetrazolium blue chloride (TB) as a leveler for copper through-hole electroplating. Although TB and MTT both belong to the tetrazolium family, their molecular architectures differ considerably: TB bears two methoxy-phenyl substituents on the tetrazolium ring, whereas MTT carries a 4,5-dimethylthiazole moiety together with two unsubstituted phenyl groups. Such substituent differences lead to distinguishable electrochemical responses and leveling outcomes. At their respective optimized conditions, TB exhibits a relatively large convection-dependent polarization response with $\Delta\eta \approx 31$ mV, yielding a TP value of 136.7% along with preferential Cu(111) growth, whereas MTT, as demonstrated in the present study, shows a smaller $\Delta\eta \approx 19.3$ mV and affords a TP value of 96.8% accompanied by a preferential Cu(200) growth (see Sections 3.4 and 3.5). A recent comparative investigation by Yang et al. likewise reported that MTT exhibits filling characteristics distinct from those of its nitro-substituted tetrazolium analog INT, further supporting the notion that subtle substituent variations within the tetrazolium family can give rise to markedly different leveling behaviors.[29] These observations together underscore that MTT represents a distinct leveler within the tetrazolium family and warrants dedicated investigation for high-aspect-ratio PCB through-hole electroplating applications.

This work systematically investigates the application performance of thiazolyl blue tetrazolium bromide (MTT) as a highly efficient leveler in PCB TH electroplating. Utilizing scanning electron microscopy and laser confocal microscopy, the exceptional microscopic leveling effect of MTT on the surface morphology of the electrodeposited copper layer is intuitively revealed. Furthermore, X-ray diffraction and contact angle measurements are employed to deeply analyze its regulatory effects on crystallographic phase evolution and solid−liquid interfacial wettability. More importantly, this study establishes a multiscale analytical framework integrating macroscopic electrochemical analyses with microscopic quantum chemical calculations to comprehensively elucidate the TH filling mechanism of MTT under multiphysics coupling. Ultimately, through precise optimization of the additive concentration ratio and related processing parameters, high-quality conformal electrodeposition in high-aspect-ratio PCB THs is successfully realized.

https://doi.org/10.1021/acs.langmuir.6c01870
Langmuir 2026, 42, 15236−15247

2. EXPERIMENTAL SECTION

2.1. Chemicals

All chemical reagents employed in this work, including copper sulfate pentahydrate ($CuSO_4·5H_2O$) and sulfuric acid (H_2SO_4, 98 wt %), were purchased from Sinopharm Chemical Reagent Co., Ltd., China. The accelerator bis(sodium sulfopropyl) disulfide (SPS, CAS No. 27206-35-5), the suppressor polyethylene glycol (PEG, molecular weight = 20,000, CAS No. 25322-68-3), and the leveler Thiazolyl Blue (MTT, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium, CAS No. 298-93-1) were all purchased from Shanghai Aladdin Biochemical Technology Co., Ltd., China. All chemicals used were of analytical grade purity (AR) and applied without any further purification process, and all test aqueous solutions were prepared using deionized water.

2.2. Electrochemical Tests

All electrochemical measurements were performed in a standard three-electrode electrolysis cell, with test control and data collection accomplished using a Gamry Reference 3000 potentiostat/galvanostat (Gamry Instruments, USA). A 5 mm-diameter platinum disk rotating electrode (RDE) was selected as the working electrode (WE). A platinum sheet electrode (1 × 2 cm^2) was used as the counter electrode, and a saturated mercurous sulfate electrode (SSE, Hg/Hg_2SO_4) served as the reference electrode (RE). Cyclic voltammetry (CV) measurements were conducted over a potential window of −0.8 to 0.2 V (vs SSE). Prior to electrochemical impedance spectroscopy (EIS) characterization, a copper layer was electrodeposited onto the RDE surface in the virgin makeup solution (VMS) at a constant current density of 1.5 A dm^{-2} for 60 s, thus converting the Pt-RDE to a Cu-RDE. EIS measurements were executed at 1000 rpm with an alternating current (AC) amplitude of 10 mV, a direct current (DC) bias potential of −0.5 V (vs SSE), and a frequency sweep range of 10^5 to 10^{-2} Hz. For chronopotentiometry (CP) tests, a constant current density of 2 A dm^{-2} was applied to the electrode system. Under normal conditions, the rate of cathodic copper electrodeposition is positively correlated with the forced convection intensity of the plating bath because forced convection reduces the thickness of the Cu^{2+} diffusion boundary layer. However, the plating bath contains not only Cu^{2+} but also various additives, which greatly complicate the mass transfer process at the cathode surface. According to the convection-dependent adsorption mechanism (CDA), a qualified plating bath system should exhibit the following behavior: a faster copper deposition rate under weakly forced convection and a relatively slower rate under strongly forced convection.[30]

2.3. Characterization

The microstructure of the electrodeposited copper layers was systematically characterized via multidimensional analytical approaches. A field-emission scanning electron microscope (Zeiss Supra55, Germany) was utilized to observe the surface morphology under an acceleration voltage of 5 kV. The surface roughness parameter (Sa) was quantitatively analyzed using an ultra-depth-of-field 3D microscope (Olympus DSX510), and the average roughness (Sa) of each sample was calculated as the mean value of measurements obtained from three different regions. X-ray diffraction (XRD, Bruker D8) was employed to investigate the preferential crystal plane orientation of the electrodeposited copper layers, and the measurements were performed at a voltage of 40 kV and a current of 40 mA, with continuous scanning at a rate of 0.02°/s over a 2θ range of 20−90°. An optical contact angle meter (Dataphysics-OCA20) was adopted to evaluate the interfacial properties, where 4 μL droplets were used for each test, and the average value was obtained from measurements conducted at five different positions.

2.4. DFT Calculations

The structural optimization of MTT was accomplished using Gaussian 09 with the B3LYP/6-311++G basis set.[31,32] The interactions between Cu^{2+} and MTT were analyzed via Multiwfn based on the Atoms in Molecules (AIM) theory and the Interaction Region Indicator (IRI) method.[33,34] Meanwhile, the intrinsic chemical properties of MTT were elucidated by means of Multiwfn,[35] combined with frontier molecular orbital theory and electrostatic potential analysis. All visualization results were rendered and output using the VMD 1.9.3 program.[36] The adsorption behavior of MTT molecules on the Cu(111) surface was investigated through calculations performed with the Vienna Ab initio Simulation Package (VASP).[37,38] For structural optimization, the generalized gradient approximation (GGA) was adopted as the exchange−correlation functional, which was parametrized using the Perdew−Burke−Ernzerhof (PBE) functional.[39] Ion-electron interactions were described and calculated by employing the projector augmented wave (PAW) pseudopotential.[40] Long-range van der Waals contributions to the MTT−Cu(111) interaction were captured through the Grimme DFT-D3 dispersion correction with zero damping (IVDW = 11). The adsorption energy of MTT on the copper substrate was calculated based on the following equation:

$$\Delta E = E_{Cu+MTT} - E_{MTT} - E_{Cu} \quad (1)$$

where E_{Cu+MTT} denotes the total energy and the energies of the optimized MTT and copper substrate are denoted by E_{MTT} and E_{Cu}, respectively.

2.5. Electroplating

The virgin makeup solution was composed of 75 g L^{-1} $CuSO_4·5H_2O$, 240 g L^{-1} H_2SO_4, and 60 mg L^{-1} chloride ions. Table 1 presents the detailed composition of the plating bath as well as the optimized process parameters selected through experimental screening. During the electroplating process, two phosphorus−copper plates were employed as the anodes, while the through-holes test coupon served as the cathode, and prior to electroplating, the test coupon was degreased with acetone under ultrasonic conditions, followed by activation treatment using a 10 wt % sulfuric acid solution and subsequent immersion in a Hull cell for electrodeposition. After the completion of electroplating, the through-holes were observed by an Olympus ultra-depth-of-field 3D microscope (Olympus DSX-510). A detailed flowchart of the experimental procedures is presented in Figure 1a.

Table 1. Bath Composition and Electroplating Process Parameters

Bath composition and process parameters	Optimum value
$CuSO_4·5H_2O$ (g/L)	75
H_2SO_4 (g/L)	240
PEG (mg/L)	250
SPS (mg/L)	1.5
MTT (mg/L)	50
Cathode current density (mA/cm^2)	20

3. RESULTS AND DISCUSSION

3.1. Surface Analysis of Electroplated Copper Layer

The regulatory effect of the MTT additive on the surface morphology of the electrodeposited copper layers was systematically investigated utilizing a scanning electron microscope and a laser confocal microscope. As depicted in Figure 1b, the copper coating electrodeposited in VMS (without MTT) exhibits a distinctly rough and uneven morphology, characterized by the presence of large and coarse crystalline particles on the surface. In sharp contrast, upon the introduction of MTT into the plating bath (Figure 1c−g), the surface flatness of the coatings shows a dramatic enhancement. With an incremental increase in MTT concentration, the crystalline morphology transitions progressively toward a highly compact and smooth state. This observation unambiguously indicates that MTT possesses

https://doi.org/10.1021/acs.langmuir.6c01870
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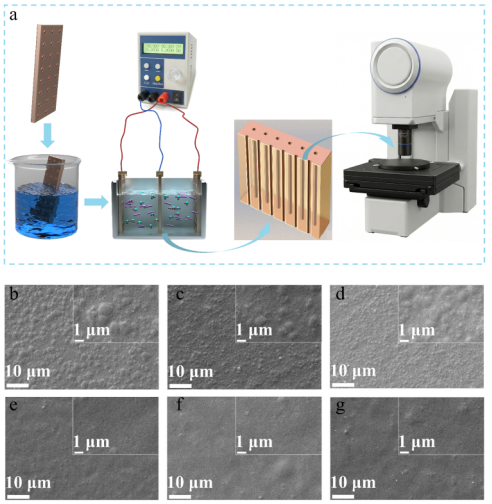


Figure 1. (a) Schematic illustration of the direct current electroplating process for through-holes. (b–g) SEM images revealing the surface morphology evolution of the copper deposits, electroplated at a constant current density of 2 A/dm^2 for 80 min, as a function of MTT concentration: (b) 0 mg/L, (c) 20 mg/L, (d) 30 mg/L, (e) 40 mg/L, (f) 50 mg/L, and (g) 60 mg/L. The main images were captured at 1000× magnification (scale bar: 10 μm), and the insets in the upper-right corners were captured at 10000× magnification (scale bar: 1 μm).

excellent leveling capability at the microscopic scale, effectively refining the grain size and upgrading the surface quality. This conclusion was further substantiated by the three-dimensional (3D) surface topography mapping. As illustrated in Figures S1 and S2, the surface roughness of the deposited layers decreases precipitously with elevating MTT concentrations, dropping from 0.545 to 0.147 μm. Such a smooth surface characteristic is highly advantageous for high-frequency signal transmission in printed circuit boards (PCBs). During high-frequency signal propagation, dictated by the skin effect,[41] the skin depth of the alternating current becomes increasingly shallow at higher frequencies. When the skin depth diminishes to a scale comparable to the surface roughness of the copper traces, the high-frequency current is forced to travel along the tortuous, rugged surface contour, thereby substantially elongating the actual conductive pathway. Consequently, a copper deposit with inherently low surface roughness can substantially reduce the transmission loss of high-frequency signals. In summary, functioning as a highly efficient leveler, MTT can significantly enhance the surface flatness and compactness of the electroplated copper layers, fulfilling the stringent physical requirements for high-frequency interconnects.

3.2. Electrochemical Analysis

Cyclic voltammetry (CV) is widely employed to preliminarily evaluate the inhibitive efficacy of additives during electrodeposition.[42] As illustrated in Figure 2a, the cathodic onset potential for copper deposition in the virgin makeup solution (VMS) without MTT is approximately −0.43 V. In contrast, the introduction of merely 20 mg/L MTT significantly intensifies the cathodic polarization, causing the onset potential to shift negatively to −0.61 V. With a further increase in MTT concentration, the cathodic overpotential continues to escalate; when the MTT concentration reaches 60 mg/L, the onset potential further decreases to −0.67 V. This evolutionary trend is highly consistent with the linear sweep voltammetry (LSV) results in Figure 1c, robustly confirming that MTT exerts a severe suppressing effect on the copper electrodeposition process. Notably, once the MTT concentration is increased to 50 mg/L, the increment in overpotential driven by concentration elevation approaches saturation, indicating that the surface adsorption and the corresponding inhibitive effect of MTT have been maximized. Furthermore, the charge density derived from the integrated area of the anodic stripping peak in the CV curves can be employed as a reliable indicator for the quantitative assessment of the inhibition performance of MTT. Figure 2b reveals that the anodic stripping coulomb of VMS is 121.86 C/dm^2, which drastically decreases to 91.37 C/dm^2 upon the addition of 20 mg/L MTT. When the concentration is elevated to 60 mg/L, the stripping coulomb plummets to 32.62 C/dm^2, representing a 73.2% reduction compared to that of VMS. This remarkable decrease in anodic stripping coulomb further corroborates the profound suppression of copper deposition induced by MTT.[43,44]

To deeply investigate the interfacial suppression mechanism of MTT, electrochemical impedance spectroscopy (EIS) analysis was conducted. The radius of the impedance semicircle in the Nyquist plot intuitively reflects the resistance of charge transfer kinetics; a larger radius signifies a more difficult interfacial charge transfer process. The impedance data were fitted using an R_u (CPE-R_{ct}) equivalent circuit model, and the extracted parameters are summarized in Table S1. R_u represents the uncompensated solution resistance, reflecting the ohmic resistance of the bulk electrolyte. As shown in Figure 2d and e, R_u remains essentially constant across all MTT concentrations, confirming that MTT exerts negligible influence on the ionic conductivity of the plating bath. R_{ct} denotes the charge transfer resistance, representing the kinetic barrier for the Cu^{2+} reduction reaction at the electrode–electrolyte interface. With increasing MTT concentration, R_{ct} rises dramatically from 24.46 Ω (0 mg/L) to 181.79 Ω (60 mg/L), which directly evidences that the adsorbed MTT layer severely impedes the interfacial electron transfer process. A constant phase element (CPE) is employed in lieu of an ideal double-layer capacitor to account for the frequency dispersion caused by surface heterogeneity, and its impedance is defined as $Z_{CPE} = 1/[Y_o(j\omega)^n]$, where Y_o is the CPE coefficient and n is the CPE exponent ($n = 1$ for an ideal capacitor). Notably, the n value decreases from 0.842 (VMS) to approximately 0.77–0.80 upon MTT addition (as shown in Table S1), suggesting that the adsorbed MTT molecules introduce additional interfacial heterogeneity, which is consistent with the formation of a nonuniform organic adsorption layer that modifies the local dielectric environment at the electrode surface.[45,46] In summary, MTT effectively suppresses the reductive deposition of copper ions.

Chronopotentiometry (CP) measurements were employed to study the convection-dependent adsorption behavior of MTT,[2,16] utilizing a low rotation speed of 100 rpm and a high rotation speed of 1000 rpm to simulate the convection intensity at the center of the through-hole and the board surface, respectively (Figure 2g). With the continuous injection of MTT, the working electrode potential exhibits a continuous negative shift, indicating that the suppression of copper electroreduction by MTT strengthens with increasing concentration, which echoes the CV and EIS findings. Moreover, the potential drop at 1000 rpm is distinctly more rapid than that at 100 rpm, revealing that the adsorption of MTT on the

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Langmuir 2026, 42, 15236−15247

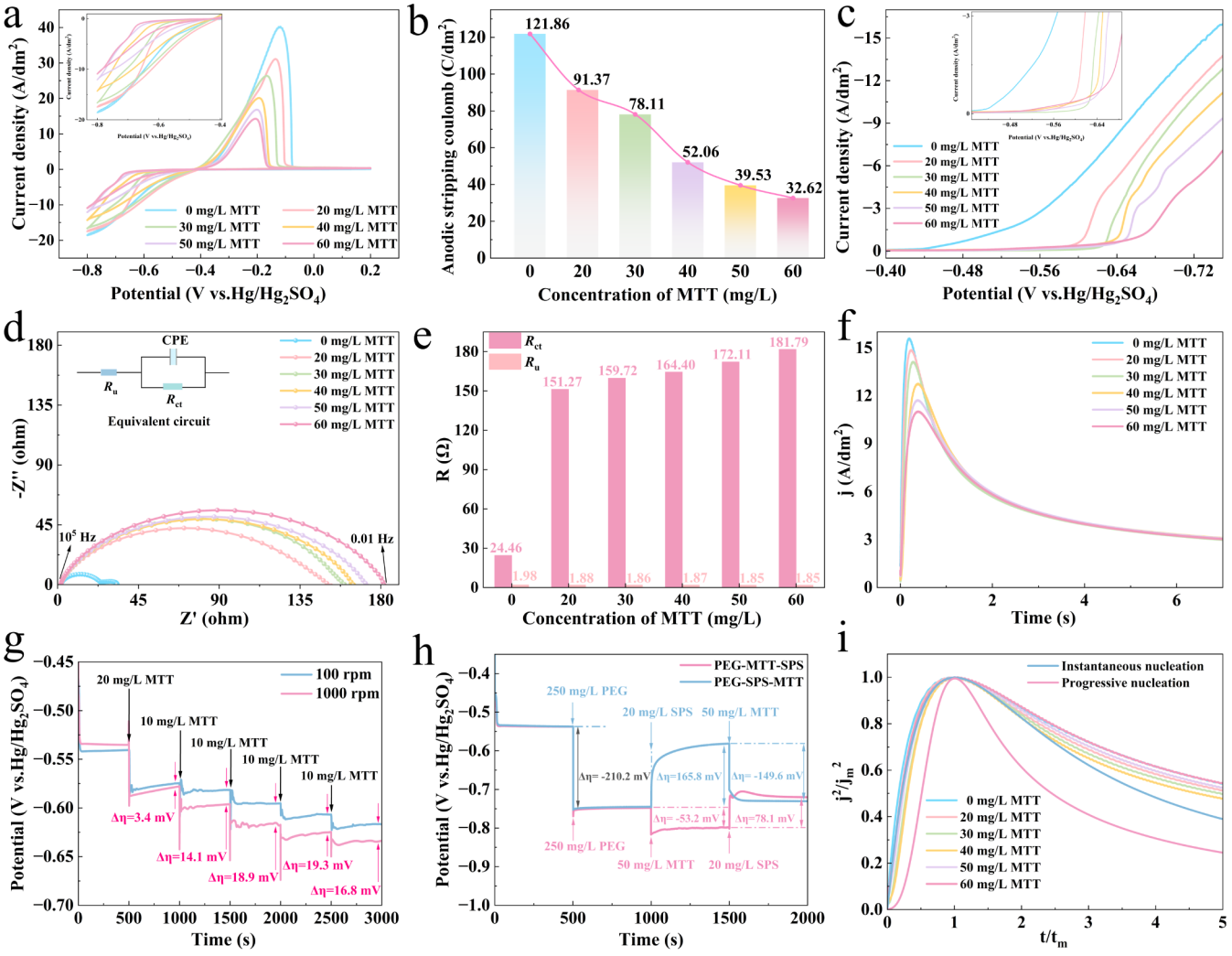


Figure 2. Electrochemical characterization of the MTT additive in the copper electroplating bath. (a) Cyclic voltammetry curves. (b) Anodic stripping charge densities calculated from the integrated areas of the anodic peaks as a function of MTT concentration. (c) Linear sweep voltammetry curves. (d) Nyquist plot. (e) Bar chart of R_{ct} and R_u. (f) Chronoamperometric current−time transients. (g) Chronopotentiometric curves recorded under varying convection intensities. (h) CP curves illustrating the interactive effects under different injection sequences of PEG, SPS, and MTT. (i) Comparison of the normalized experimental CA transients with the theoretical dimensionless curves for 3D instantaneous and progressive nucleation based on the Scharifker−Hills (SH) model.

electrode surface is predominantly diffusion-controlled. The potential difference ($\Delta\eta$) between 100 and 1000 rpm under various MTT concentrations can be calculated to evaluate the microvia filling performance; typically, a larger $\Delta\eta$ is more conducive to achieving excellent superfilling. Figure 2g demonstrates that as the MTT concentration increases from 20 mg/L to 50 mg/L, $\Delta\eta$ broadens significantly from 3.4 mV to 19.3 mV. However, with a further increase in MTT concentration, $\Delta\eta$ follows a descending trend. The relatively high $\Delta\eta$ values maintained within the concentration window of 40−50 mg/L suggest that this range is optimal for the conformal filling of through-holes. It is noteworthy that the $\Delta\eta$ values obtained from the CP measurements are well correlated with the actual through-hole filling results. A larger $\Delta\eta$ signifies a greater polarization disparity between the high-convection region (hole opening) and the low-convection region (hole center), which serves as the electrochemical driving force for redistributing the deposition current toward the hole interior.[16,30,47] The optimal $\Delta\eta$ window of 40−50 mg/L coincides precisely with the MTT concentration that yields the highest throwing power (TP = 96.8% at 50 mg/L, as will be shown in Section 3.5), thereby confirming the predictive validity of the CP-based evaluation for through-hole filling performance. Additionally, chronopotentiometry is frequently utilized to probe the synergistic or competitive interactions among multicomponent additives in the plating bath. By altering the injection sequence of the three core additives, the dynamic interactive mechanisms among MTT, bis(3-sulfopropyl)-disulfide (SPS), and polyethylene glycol (PEG) were systematically analyzed. As delineated by the blue line in Figure 2h, the sequential introduction of SPS and MTT into a system predosed with PEG triggers intense depolarization (a positive shift of 165.8 mV) and polarization (a negative shift of 149.6 mV) responses, respectively. Similarly, the pink line in Figure 2h indicates that the direct addition of MTT following PEG induces a further increase of cathodic polarization (−53.2 mV), verifying a pronounced synergistic inhibitory effect between MTT and PEG. Comparing the two potential evolution curves reveals that despite the different addition pathways, the final steady-state potentials of the systems

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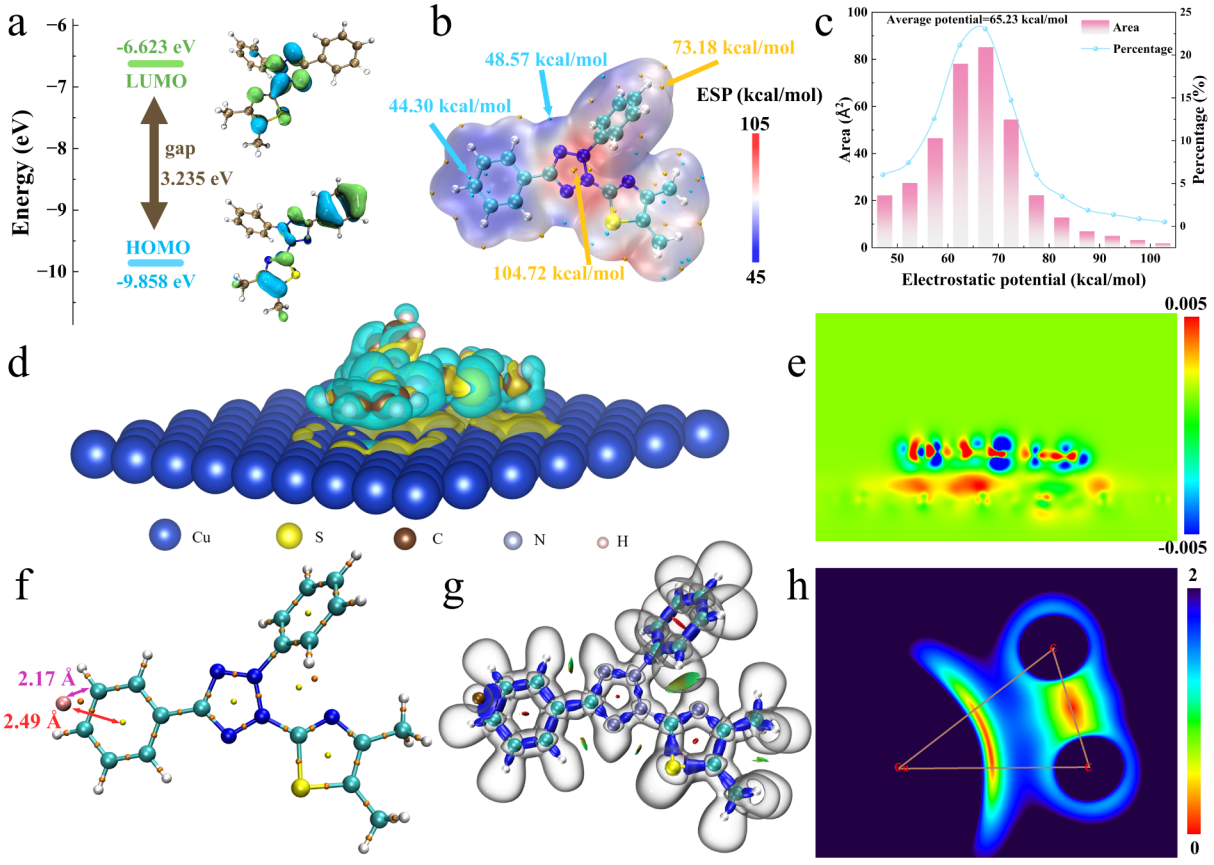


Figure 3. (a) Frontier molecular orbitals (HOMO and LUMO) of the MTT molecule. (b) Electrostatic potential surface map and (c) the corresponding quantitative distribution histogram of the ESP. (d) Three-dimensional (3D) and (e) two-dimensional (2D) Bader charge density difference maps of the MTT molecule adsorbed on the copper substrate (red regions represent electron accumulation, and blue regions represent electron depletion). (f) AIM topological analysis map of the MTT−Cu^{2+} cation−π interaction. (g) 3D isosurface map and (h) 2D color-mapped plane projection of IRI for the MTT−Cu^{2+} cation−π configuration (green regions: van der Waals interactions; blue regions: attractive noncovalent π-system perturbations associated with the cation−π contact).

remain virtually identical. This demonstrates that MTT and the accelerator SPS undergo competitive adsorption on the cathode surface, thereby jointly regulating the electrodeposition morphology of copper.

Finally, chronoamperometry (CA) was employed to investigate the influence of MTT on the initial nucleation and growth kinetics of copper (Figure 1f). Under a potential step, the current density rapidly increases to a maximum value (j_m) within a very short period and subsequently declines to a steady state due to diffusion-limited mass transfer. The initial current transient corresponds to the formation of nuclei and 3D growth during electrodeposition. According to the standard Scharifker−Hills (SH) 3D nucleation theoretical model, the instantaneous and progressive nucleation processes under 3D diffusion control obey the following dimensionless equations, respectively:

For instantaneous nucleation:

$$(j/j_m)^2 = \frac{1.9542}{t/t_m}\{1 - \exp[-1.2564(t/t_m)]\}^2 \quad (2)$$

For progressive nucleation:

$$(j/j_m)^2 = \frac{1.2254}{t/t_m}\{1 - \exp[-2.3367(t/t_m)^2]\}^2 \quad (3)$$

where j is the maximum current density and j_m is the time required to reach the peak current density. On the basis of the rigorous SH theoretical model, the experimentally acquired current−time transients were normalized (Figure 2i). The results indicate that, regardless of the presence or absence of MTT in the plating bath, the dimensionless experimental data points closely align with the theoretical curve for 3D instantaneous nucleation. This unambiguously indicates that, while MTT alters the kinetic rate of electrodeposition, it exerts a negligible impact on the inherent nucleation mechanism and growth mode of the copper.

3.3. Theoretical Insights into the Inhibition Mechanism

In this section, Density Functional Theory (DFT) calculations were conducted to unravel the interfacial adsorption and coordination effects of MTT during copper electrodeposition, thereby systematically elucidating its inhibition mechanism at the microscopic scale. The optimized geometric configuration of the MTT molecule is depicted in Figure S3. Frontier Molecular Orbital theory is extensively employed to analyze and predict the interfacial adsorption behavior of molecules. As illustrated in Figure 3a, the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) are predominantly localized on the tetrazole and thiazole rings, indicating that these functional groups serve as active centers for preferential adsorption.[10,48] Furthermore, the surface electrostatic potential distribution histogram (Figure 3c) reveals that the overall MTT molecule is positively charged,[49,50] possessing a remarkably high average surface

https://doi.org/10.1021/acs.langmuir.6c01870
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Table 2. AIM Analysis-Derived Electron Density (ρ_{BCP}), Laplacian of Electron Density ($\Delta^2\rho_{BCP}$), Energy Density (H_{BCP}), and Potential Energy Density (Vr_{BCP}) at the Bond Critical Point

Coordination configuration	Interaction	$\rho_{BCP} \times 10^2$ (au)	$\nabla^2\rho_{BCP} \times 10^2$ (au)	$H_{BCP} \times 10^3$ (au)	$Vr_{BCP} \times 10^3$ (au)
MTT-Cu^{2+}	Ph···Cu^{2+}	5.68	23.47	−19.97	−62.67

electrostatic potential of 65.23 kcal/mol. Driven by robust electrostatic attraction during the electrodeposition process, MTT is highly prone to adsorbing onto the cathodic surface, thereby effectively suppressing copper reduction. To further probe the interfacial adsorption dynamics of MTT on the copper substrate, charge density difference calculations were performed to quantitatively assess its adsorption behavior (Figure 3d and e). The pronounced interfacial charge redistribution between the copper substrate and the MTT molecule decisively confirms their close geometric proximity and intense electronic interaction. Further thermodynamic calculations yield a highly negative adsorption energy of −2.93 eV. This value is consistent with—and slightly more conservative than—the magnitude reported for MTT on Cu(111) under an essentially equivalent level of theory (PBE + Grimme D3 dispersion, slab calculation: $E_{ads} \approx -8.14$ eV; −785.93 kJ mol^{-1}) by Yang et al.,[29] confirming that the strong MTT−Cu(111) interfacial adsorption is robustly captured by the present calculation. Collectively, these quantum chemical analyses substantiate that MTT can robustly adsorb onto the cathode surface during electroplating.

To deeply investigate the coordination interaction between MTT and Cu^{2+}, a multidimensional analysis integrating electrostatic potential (ESP) mapping, Local Electron Affinity (LEA), Atoms in Molecules (AIM) theory, and the Interaction Region Indicator (IRI) was employed. Because MTT exists in solution as a tetrazolium cation, its molecular surface is positive throughout (Figure 3b,c; average ESP ≈ 65.23 kcal/mol), and the relevant descriptor governing the site at which Cu^{2+} is preferentially captured is therefore not the (still positive) local ESP minimum but rather the local ionization energy and the polarizability of the underlying electron density. The Average Local Ionization Energy (ALIE, Figure S4) reaches its lowest value in the vicinity of a phenyl-ring carbon, identifying this position as the most readily polarizable, electron-donating site of the molecule. Coupled with the high π-electron density of the phenyl ring, this site is electronically primed to bind Cu^{2+} via a cation−π interaction, in which the attractive force originates from polarization of the π-cloud by the divalent cation rather than from a permanent electrostatic well.[51,52] Figure 3f displays the topological (AIM) analysis of the MTT−Cu^{2+} cation−π configuration. In the converged Gaussian geometry, Cu^{2+} resides 2.07 Å above the phenyl-ring plane, with a Cu^{2+}···ring-centroid distance of 2.49 Å. The cation is offset 1.39 Å from the centroid axis so that the two shortest Cu^{2+}···C contacts are with the meta- and para-carbons of the ring (2.18 Å and 2.23 Å, respectively). This off-centroid, η^2-like geometry is the canonical configuration of cation−π complexes formed by open-shell d^9 transition-metal cations such as Cu^{2+}, which preferentially bind in an asymmetric fashion to two adjacent ring carbons rather than over the ring centroid.[51,52] The physicochemical nature of the interaction was quantitatively evaluated by extracting the topological parameters at the bond critical point (BCP, Table 2). The electron density ρ_{BCP} = 0.0568 au (<0.1 au), the positive Laplacian $\nabla^2\rho_{BCP}$ = 0.2347 au, and the slightly negative total energy density H_{BCP} = −0.01997 au with Vr_{BCP} = −0.06267 au together constitute the textbook closed-shell, noncovalent signature of a cation−π interaction; the modest negative H_{BCP} further indicates a partial covalent stabilization, which is characteristic of cation−π contacts involving open-shell d-block metal cations.[53] This DFT-derived picture is qualitatively supported by the UV−vis nonadditivity measurements presented in Figure S7, which evidence a localized perturbation of the Cu^{2+} d−d band—consistent with the partial displacement of an axially bound water from the $[Cu(H_2O)_6]^{2+}$ first coordination sphere that necessarily accompanies the formation of a cation−π contact in solution. Moreover, as observed in Figure 3a, the energy gap of the MTT molecule is merely 3.235 eV. Such a narrow orbital bandgap implies that MTT readily undergoes electron transition and charge transfer with copper ions, which is in seamless agreement with the AIM topological analysis. The spatial interaction between MTT and Cu^{2+} was visually conceptualized via the IRI method (Figure 3g and h). In the isosurface map (Figure 3g), the distinctive, continuous blue isosurface emerging between MTT and Cu^{2+} unequivocally proves a cation−π interaction, a phenomenon that is also intuitively visualized in the 2D color-mapped plane plot (Figure 3h).

From the above analyses, MTT can spontaneously adsorb onto the cathode interface in the plating bath, while simultaneously interacting with bulk Cu^{2+} ions. To confirm the dominant role of interfacial adsorption, in situ FTIR spectra of the Cu working electrode were collected in VMS electrolyte containing 50 mg L^{-1} MTT at applied potentials ranging from −0.45 V to −0.65 V (vs SSE, 50 mV step). As shown in Figure S8, a series of vibrational features in the 1000−1700 cm^{-1} fingerprint region—attributable to the phenyl, tetrazolium, thiazole, and methyl moieties of MTT (phenyl and tetrazolium C═C/C═N stretches near 1500−1620 cm^{-1}, thiazole/phenyl ring deformations near 1400−1500 cm^{-1}, methyl C−H bending near 1380−1410 cm^{-1}, C−N stretches near 1270−1330 cm^{-1}, and ring-breathing modes near 1040−1150 cm^{-1})—grow monotonically in intensity as the applied potential becomes more cathodic, providing direct in situ evidence for progressive MTT accumulation at the Cu cathode interface. Since the bulk cation−π interaction is much weaker than interfacial adsorption, it cannot appreciably hinder the diffusion and subsequent adsorption of MTT at the cathode. Consequently, the overall inhibition efficiency is governed primarily by interfacial adsorption.

Based on the aforementioned theoretical calculations, topological analyses, and in situ FTIR spectroelectrochemical measurements, it can be concluded that the inhibition mechanism of MTT in the electrodeposition system arises from an asymmetric combination of two effects of markedly different magnitudes. At the cathode interface, MTT adsorbs strongly to form a dense blocking molecular layer that reduces the available active sites for Cu^{2+} reduction (interfacial-adsorption effect; the dominant kinetic contribution). In the bulk electrolyte, a small fraction of MTT engages Cu^{2+} ions through a comparatively weak cation−π interaction, modestly raising the activation barrier for Cu^{2+} reduction at the eventual

https://doi.org/10.1021/acs.langmuir.6c01870
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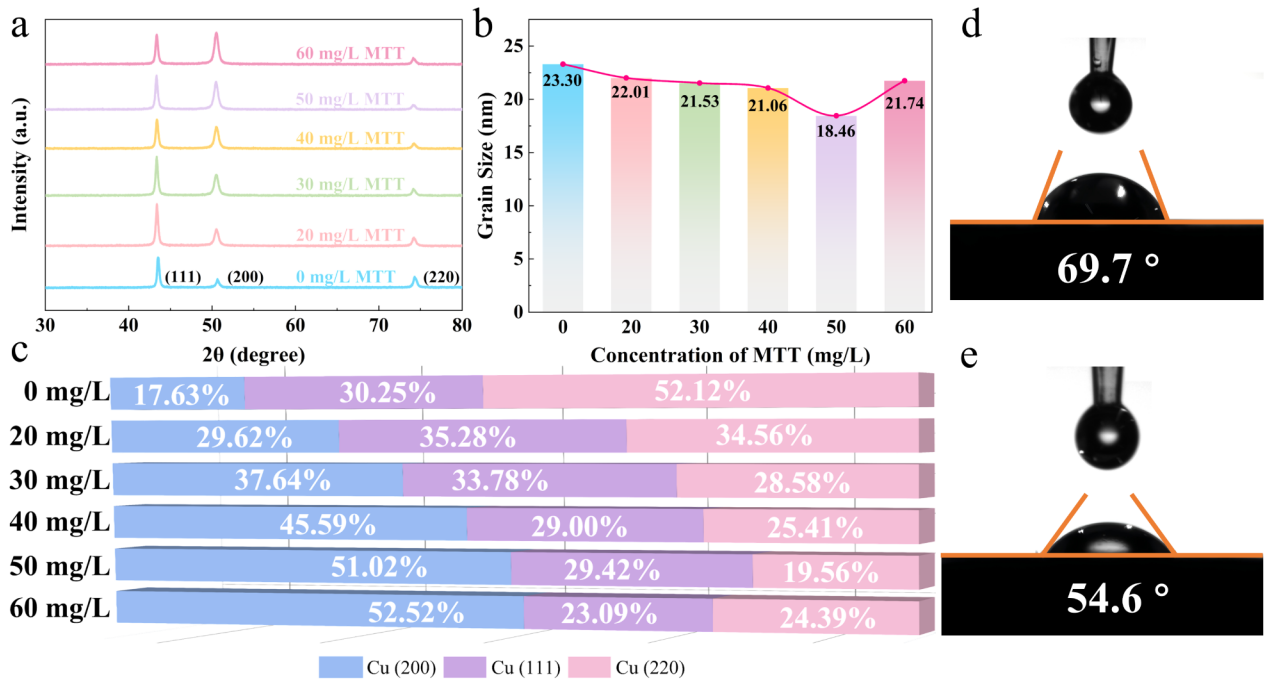


Figure 4. (a) XRD patterns of the copper layers electrodeposited from baths containing various concentrations of the MTT additive. (b) Average grain size and (c) relative texture coefficients ($RTC_{(hkl)}$) of the corresponding copper deposits as a function of MTT concentration. Contact angle measurements on the PCB surface using different plating solutions: (d) VMS and (e) VMS + 50 mg/L MTT.

interface (bulk cation−π contribution; a minor secondary effect).

3.4. Phase Structure and Interface Analysis of Electroplated Copper Layer

The impact of MTT on the crystallographic structure and grain size evolution of the electroplated copper layers was investigated utilizing XRD. According to the XRD patterns (Figure 4a), characteristic diffraction peaks corresponding to the (111), (200), and (220) planes of copper are distinctly observed at 2θ values of 43.3°, 50.4°, and 74.1°, respectively. As depicted in Figure 4a, with the incremental addition of MTT into the plating bath, the intensity of the (111) peak exhibits no obvious variation, whereas the (200) peak progressively intensifies and the (220) peak gradually attenuates. To quantitatively elucidate the regulatory effect of MTT on the preferred crystallographic orientation, the relative texture coefficients ($RTC_{(hkl)}$) for the respective lattice planes were calculated based on the following equation:

$$RTC_{(hkl)} = \frac{I_{(hkl)}/I_{0(hkl)}}{\sum (I_{(hkl)}/I_{0(hkl)})} \times 100\% \quad (4)$$

where $I_{0(hkl)}$ represents the standard diffraction peak intensity of a randomly oriented copper powder sample and $I_{(hkl)}$ denotes the measured diffraction peak intensity of the actual electrodeposited copper films. The quantitative results (Figure 4c) reveal that as the MTT concentration escalates from 0 mg/L to 60 mg/L, $RTC_{(200)}$ sharply surges from 17.63% to 52.52%, $RTC_{(111)}$ exhibits a marginal decline, and $RTC_{(220)}$ plummets dramatically from 52.12% to 24.39%. This rigorous texture coefficient analysis unambiguously demonstrates that the incorporation of MTT does not induce a preferred orientation along the Cu(111) plane, but rather intensely promotes the preferential growth of the Cu(200) plane. Furthermore, the Scherrer equation was employed to quantitatively assess the influence of MTT concentration on the crystalline grain size (D) of the copper deposits:

$$D = \frac{K\lambda}{\beta \cos \theta} \quad (5)$$

where K is the Scherrer constant (assigned as 0.89), λ is the incident X-ray wavelength (0.1541 nm), β represents the full width at half-maximum (fwhm) of the diffraction peak, and θ is the corresponding Bragg diffraction angle. The calculated results (Figure 4b) dictate that within the concentration range of 0−50 mg/L, the grain size is notably refined from 23.30 to 18.46 nm as the MTT concentration elevates. Nevertheless, further exceeding this MTT concentration threshold leads to an anomalous grain coarsening. This phenomenon indicates that there exists an optimal concentration window for MTT, within which the most highly efficient grain refinement can be achieved.

Although the orientation evolution analyzed above and the conformal filling reported in Section 3.5 are governed by different proximate mechanisms—the former by the facet-selective character of MTT adsorption on the cathode and the latter by the convection-dependent adsorption (CDA)—they share a common origin in the surface adsorption of MTT and concurrently emerge at the same optimum concentration (50 mg/L). The two effects are functionally complementary in the present context: CDA establishes uniform Cu thickness from the orifice to the via center, while the Cu(200) preferential growth and the grain refinement to 18.46 nm provide a fine-grained, densely packed microstructure throughout the conformal layer. The resulting deposit therefore satisfies simultaneously the two principal requirements of PCB through-hole metallization, namely conformal thickness uniformity (which guarantees interconnect reliability) and low surface roughness (which mitigates skin-effect-induced signal losses).

https://doi.org/10.1021/acs.langmuir.6c01870
Langmuir 2026, 42, 15236−15247

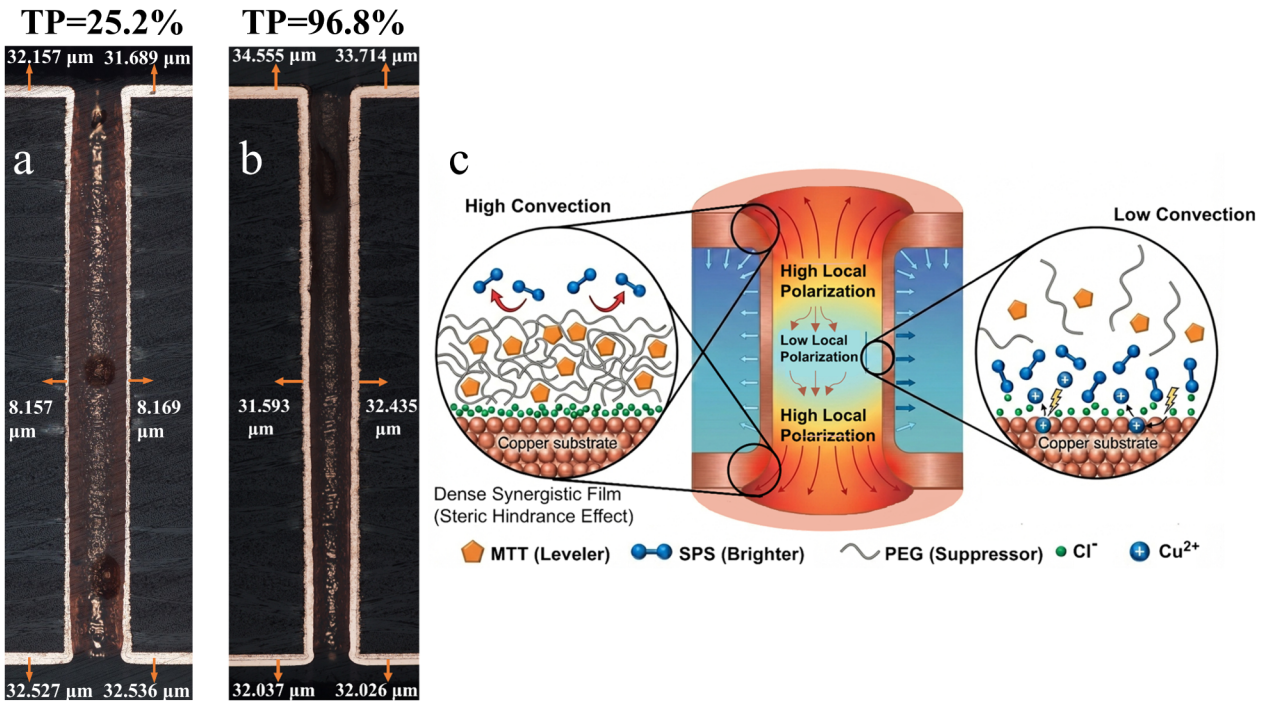


Figure 5. Metallographic cross-sectional optical micrographs of PCB through-holes electroplated in (a) VMS + 1.5 mg/L SPS + 250 mg/L PEG and (b) VMS + 1.5 mg/L SPS + 250 mg/L PEG + 50 mg/L MTT. (c) Schematic illustration of the proposed mechanism for the conformal through-hole electrodeposition facilitated by MTT.

Finally, contact angle measurements were conducted to evaluate the wettability modulation of MTT on the PCB substrates. During the electrodeposition process, a plating bath endowed with superior wettability can significantly lower the interfacial surface tension. This effectively weakens the adhesion of microscopic hydrogen bubbles (generated by side reactions) to the electrode surface, promoting their prompt detachment. Consequently, the formation of macroscopic defects such as pinholes and pitting on the coating surface is fundamentally suppressed. As illustrated in Figure 4d, the initial contact angle of the virgin makeup solution (VMS) on the PCB is 69.7°. Strikingly, upon the addition of 50 mg/L MTT, the contact angle decreases to 54.6° (Figure 4e). This distinct decrease in the contact angle verifies that the introduction of MTT substantially upgrades the overall wettability of the plating bath toward the substrate.

3.5. TH Electrodeposition Performance

To comprehensively evaluate the through-hole filling capability of MTT, the thickness uniformity of its electrodeposited copper layers within printed circuit board (PCB) through-holes was systematically characterized. In the PCB through-hole electroplating process, Throwing Power (TP) serves as the pivotal quantitative index for evaluating the internal deposit thickness uniformity and evaluating the quality of the copper interconnects. The magnitude of the TP directly dictates the electrical reliability and long-term service lifespan of the 3D interconnected structures. Figures 5a,b, and S6 display high-resolution metallographic cross-sectional optical micrographs of the through-holes after electrodeposition under various MTT concentrations. At a constant current density of 2 A/dm^2, when VMS exclusively contains the accelerator SPS and the suppressor PEG (Figure 5a), the TP value of the through-hole is only 25.2%. At this juncture, a distinct geometric disparity exists between the deposit thickness at the opening and that in the central region, resulting in a severe "dog-bone" morphological defect that fundamentally fails to meet the electrical reliability requirements for high-density interconnects. Remarkably, with the introduction of MTT into the plating system, this geometric effect of uneven deposit distribution is effectively suppressed, and the overall coating morphology transits almost into a conformal deposition. As illustrated in Figure 5b, upon adding an optimal MTT concentration of 50 mg/L, the TP value increases to 96.8%. This visually dictates that MTT bridges the thickness gap between the surface opening and the deep interior, achieving high-quality conformal thickening of the copper layer within the through-hole.

To further benchmark the leveling performance of MTT against a well-established commercial leveler, a comparative through-hole electroplating experiment was conducted using 50 mg/L Janus Green B (JGB) under identical bath composition and process conditions. As illustrated in Figure S6d, the cross-sectional metallographic micrograph of the through-hole electroplated with JGB reveals a relatively uniform copper distribution along the hole wall, with no pronounced "dog-bone" defect observable at the orifice. Nevertheless, the overall copper deposit thickness within the through-hole remains noticeably thin compared to that achieved with MTT, yielding a TP value of approximately 84.5%. Although this TP value represents a substantial improvement over the additive-free system (TP = 25.2%), it remains significantly inferior to the 96.8% attained with 50 mg/L MTT. This performance disparity can be rationalized by the stronger cathodic polarization and more pronounced convection-dependent adsorption behavior of MTT, as demonstrated by the electrochemical analyses in Section 3.2. The higher inhibitory potency of MTT at the forced-convection orifice region enables a more effective redistrib-

https://doi.org/10.1021/acs.langmuir.6c01870
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ution of the electric field lines toward the through-hole interior, thereby yielding superior thickness uniformity. These comparative results unequivocally corroborate the exceptional leveling efficacy of MTT and its marked advantage over the conventional JGB leveler in high-aspect-ratio through-hole electroplating applications.

Coupling the theoretical elucidation of the MTT inhibition mechanism in Section 3.3 with previous studies, competitive interfacial polarization remodeling driven by a multiphysics coupling mechanism was proposed to unveil the intrinsic influence of MTT. As portrayed in the schematic mechanism (Figure 5c), governed by the coupling of fluid dynamics (convective mass transfer) and electrochemical kinetics, MTT, PEG, and SPS undergo competitive adsorption at the solid−liquid interface within the through-hole. This induces a spatially differentiated distribution of polarization resistance, reshaping the potential field across the inner surface of the through-hole. Specifically, propelled by the forced convection at the hole opening, MTT rapidly accumulates and adsorbs in this region via convection-dependent adsorption (CDA).[54,55] Concurrently, MTT interacts synergistically with the PEG polymer chains and the bridging chloride ions to form a dense ternary synergistic suppression film on the copper substrate. This compact adsorption layer triggers a notable increase in the local polarization resistance at the opening. The resulting elevated electrochemical reaction barrier effectively suppresses the electroreduction kinetics of Cu^{2+} in this vicinity. Consequently, the increased polarization resistance at the orifice compels the spatial electric field lines (i.e., the primary current distribution) to redistribute toward the deep interior of the through-hole, which is situated in a weak convection zone.[56,57] Through this polarization remodeling effect, the inherent Ohmic potential drop caused by the geometric aspect ratio of the through-hole is compensated. Simultaneously, a steady-state diffusion flux of Cu^{2+} directed into the hole is sustained, ultimately driving the realization of a conformal through-hole deposition.

4. CONCLUSIONS

In summary, this study systematically investigated the application performance and multiphysics coupling mechanisms of MTT, acting as a leveler for copper electrodeposition in PCB through-holes. Electrochemical evaluations demonstrate that MTT significantly enhances cathodic polarization via a convection-dependent interfacial adsorption. Meanwhile, microscopic morphological characterization confirms that MTT exhibits excellent leveling capability, significantly reducing the surface roughness of the copper deposit to 0.147 μm. Such a smooth surface topography is highly beneficial for mitigating the skin effect and lowering insertion loss during high-frequency signal transmission. Density functional theory analysis further elucidates that the microscopic origin of this robust inhibitory behavior lies primarily in the strong interfacial adsorption of MTT at the cathode, with the bulk cation−π interaction with Cu^{2+} contributing as a minor secondary effect. Phase structure and interface analysis illustrate that an optimal MTT concentration (50 mg/L) not only effectively refines the crystalline grain size and promotes the preferential growth of the Cu(200) plane but also decreases the solid−liquid interfacial contact angle to 54.6°, thereby fundamentally suppressing the generation of macroscopic morphological defects. Ultimately, in practical through-hole electroplating applications, the competitive adsorption among MTT, PEG, and SPS induces a spatially differentiated remodeling of polarization resistance. By establishing a high-impedance barrier at the forced-convection orifice region, the inherent Ohmic potential drop within the through-holes is effectively compensated. Driven by this multiphysical coupling mechanism, the TP value of through-holes is substantially elevated from 25.2% to 96.8%, markedly outperforming the conventional leveler JGB (TP = 84.5%) under identical conditions, successfully achieving high-quality conformal deposition and providing a theoretical and practical basis for the fabrication of highly reliable, high-frequency 3D interconnected PCB structures.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.langmuir.6c01870.

Summary of fitting parameters for EIS equivalent circuit; 3D surface topography mapping of electroplated copper layers at different concentrations of MTT; surface roughness histogram of electroplated copper layers at different concentrations of MTT; the optimized structure of the MTT molecule; local electron affinity map and average local ionization energy map; contact angle measurements on the PCB surface; metallographic cross-sectional optical micrographs of PCB through-holes; UV−vis spectroscopy; in-situ infrared spectroscopy; detailed descriptions regarding the adsorption energy calculation (PDF)

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https://doi.org/10.1021/acs.langmuir.6c01870
Langmuir 2026, 42, 15236−15247

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Notes

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■ ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China [No. 22541202, 22402041, 22502032]. Numerical computations were performed at the Hefei Advanced Computing Center .

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https://doi.org/10.1021/acs.langmuir.6c01870
Langmuir 2026, 42, 15236−15247